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MSW-Net: A hierarchical stacking model for automated municipal solid waste classification

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ABSTRACT

Efficient solid waste management is crucial for urban health and welfare in the midst of fast industrialization and urbanization. In this changing environment, government authorities have a significant role in addressing and reducing the effects of solid waste. While waste separation at the source simplifies processes, manual sorting is a consequence of ignorance in numerous regions, which endangers the health of waste pickers. This study addresses the challenges by introducing the MSW-Net model, a hierarchical stacking model designed for the automated classification of municipal solid waste (MSW). Customized Convolutional Neural Network (custom CNN) and Bayesian-Optimized MobileNet models serve as the base models, with Gradient Boosting employed as the meta-classifier. The MSW-Net model, as proposed, exhibits exceptional performance, attaining 99%, 95%, and 96.43% accuracy rates over training, validation, and testing, respectively. Additionally, the model achieves precision, recall, and F1 scores of 96.42%, 96.43%, and 96.42% during the testing phase. Therefore, the proposed MSW-Net model performs better than the other existing models in sorting the waste. This could also aid the municipal authorities in classifying the waste with minimal human intervention.

Implications: The MSW-Net model, featuring a hierarchical stacking approach with custom CNN and Bayesian-Optimized MobileNet base models, and Gradient Boosting as the meta-classifier, achieves remarkable accuracy in automated municipal solid waste classification. With performance metrics of 99% accuracy in training, 95% in validation, and 96.43% in testing, alongside precision, recall, and F1 scores around 96.42%, the MSW-Net model significantly outperforms existing models. This advancement promises to aid municipal authorities in efficient waste management, reducing reliance on manual sorting and thereby improving the health and safety of waste pickers.

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Introduction

The demand for commodities rises in tandem with increasing income, driving up production and consumption rates. Renowned environmentalist Jacques Cousteau stated, "Population expansion is the major source of environmental damage" (Elnadi and Rifaat 2023). As the quote suggests, increased production and consumption results in generating a significant amount of waste. The World Bank estimates that approximately 683 million tons of waste are generated per day globally (Kaza et al. 2018). Managing waste is a critical factor in maintaining a healthy and sustainable environment. Among the three major types of issues, waste management falls under Type II, Green issues, dealing with environmental concerns such as climate change, pollution, and resource depletion (Sajdl., 2005). Waste is classified into various types, including Municipal Solid Waste (MSW), agricultural waste, bio-medical waste, chemical waste, radioactive waste, and inert waste. MSW is the primary waste source which is estimated to reach 3.4 billion tons by 2050 (World Bank Group 2018). According to the Environmental Protection Agency, MSW comprises paper (27%), food scraps (14.6%), yard trimming (13.5%), plastics (12.8%), metals (9.1%), rubber, leather, textiles (9%), Wood (6.2%), glass (4.5%), and other miscellaneous materials (3.3%) (Municipal Solid Waste 2022).

Every day, India generates about 1,50,000 metric tons of MSW, with 47% processed and treated, 27% ending up in landfills, 23% remaining untreated, and 3% not collected (Biswas and Parida 2021). Collectively, more than half of waste generated in India either reaches dumpsites or remains unattended. However, India aims to eliminate litter in urban areas through programs such as Solid Waste Management Rules 2016, AMRUT, Clean India Mission 2014, and Swachh Bharat Mission 2.0 (SBM 2.0). These plans emphasize scientific waste treatment to manage both existing and estimated

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municipal waste. Additionally, SBM 2.0 mandates municipalities to clean up old garbage sites by the end of March 2024 (MoHUA, 2023). Nevertheless, the reality in major cities across the country raises concerns. There are approximately 15,000 hectares of legacy garbage in the country, roughly equivalent to 14,500 football fields. According to the Central Public Health and Environment Engineering Organization, under the Ministry of Housing and Urban Affairs, the country loses an additional 230 acres of land to MSW every year (Centre for Science and Environment 2023). Unsanitary landfills contaminate air, water, and soil.

Effectively managing waste involves six fundamental steps including segregation at the source, collection, transportation, processing, recycling, and disposal of waste materials. The initial step, segregation at the source, makes the remaining steps more convenient. However, waste is not sorted at its source in many parts of the country, requiring manual segregation after collection and transportation. India has approximately 3184 dumpsites, of which only 234 have been reclaimed or capped, and eight have been converted to sanitary landfills (Ahuja 2022; Centre for Science and Environment 2023). Waste pickers, comprising both formal and informal sectors, work hard to segregate the waste at these sites. These individuals form the largest workforce in the recycling chain and recover a higher proportion of recyclable materials than formal waste management systems. According to a 2013 report by the International Labour Organization, it was estimated that over 20 million people worldwide work as waste pickers (Rava Zolnikov 2023). This traditional method not only affects the health of people and the environment but also consumes a significant amount of time and labor force.

To address these challenges, this research introduces the MSW-Net model, a novel Deep Learning (DL) approach designed for the automated classification of municipal solid waste (MSW). The proposed model utilizes advanced image recognition techniques to enhance efficiency and accuracy in waste sorting. Custom Convolutional Neural Network (custom CNN) and Bayesian-Optimized MobileNet models serve as the base models, with Gradient Boosting employed as the meta-classifier. The novelty of the proposed MSW-Net model lies in its hierarchical structure and the synergy between DL models and mechanical separation systems. While other models have been developed based on similar principles, our approach is distinguished by its high accuracy rates. Furthermore, the use of Bayesian optimization for model tuning ensures optimal performance, setting the proposed model apart from existing solutions.

This Deep Learning approach could help authorities handle MSW more effectively by integrating this model with IoT-based waste sorting machines. Further, promoting sustainability could positively impact the environment, the economy, and public health. Classifying waste at the source without human intervention could increase the recycling rate, consequently reducing the waste disposal rate. Many fellow researchers had analyzed the MSW management using the ML and DL techniques. Some of the articles employing the diverse deep-learning models such as ResNet, MobileNet, EfficientNet, and CNN were discussed here.

A waste classification system was proposed using ResNet and MobileNet models (Yang et al. 2023). In addition, object detection was also analyzed using the three different algorithms of the YOLOv5 family. The training results of models were integrated and predicted through the consensus voting algorithm. The recognition rate of around 97.6% was achieved through the consensus voting algorithm. A computer vision system was developed by Bchir et al. to sort PET waste (Bchir et al. 2021). U-Net, Fully Connected Network (FCN), Segnet and DeepLabV3+ were employed to semantically segment the objects. The ResNet model was used as the backbone for constructing the DeepLabV3+. The proposed DeepLabV3+ model outperformed with a weighted Intersection over Union (wIoU) of 97%. An efficient MSW management system that uses computer vision and IoT devices to distinguish between biodegradable and non-biodegradable garbage was developed (Mookkaiah et al. 2022). The pre-processing method of batch normalization was performed. In addition to it, the Inception ResNetV2 derived CNN model along with a mixed hybrid pooling technology was built to enhance the performance. CNN using the ResNet model obtained accuracy of 94.4%.

The solid waste was classified using a wide variety of criteria, including 34 distinct groups (Malik et al. 2022). The transfer learning method was implemented by utilizing EfficientNetB0 to B7. The EfficientNetB3 model obtained 85% accuracy in recognizing the type of waste. Similarly, using the EfficientNet model garbage localization and classification were performed (Majchrowska et al. 2022). The EfficientNetD2 model was used to localize the waste while the EfficientNetB2 sorted the same. Both performed efficiently with accuracy of 70% and 75%. Bobulski et al. utilized the WaDaBa dataset to classify plastic waste (Bobulski and Kubanek 2021). The following models were employed AlexNet, Modified CNN, and MobileNet. Comparatively, the MobileNet model achieved a greater accuracy of 86% than the modified CNN model. The author claims that though the proposed CNN model is less effective, it has much

fewer parameters. Thus, it could have significant impacts in the case of implementation for mobile devices. NWNU trash dataset was utilized to categorize the type of waste (Zhang et al. 2021). Various DL models such as AlexNet, GoogleNetV2, VGG16 and DenseNet169 were utilized. The transfer learning based DenseNet169 model outperformed the other employed models by obtaining accuracy of 82.8%.

An ensembled model was developed to recognize the trash (Zheng and Gu 2021). The ensemble model includes three pre-trained CNN models and the Unequal Precision Management Weighting Strategy (UPMWS). The UPMWS model was used to determine the weight coefficients of the ensembled model. In combination with the UPMWS, GoogleNet, ResNet and MobileNetV2 were utilized. The ensembled CNN voting model performed the best with accuracy of 92.32%. A multi-task learning architecture (MLTA) was proposed based on a CNN model (Liang and Gu 2021). The proposed model could localize and recognize the waste type. The ResNet50 model along with an attention module and a novel feature pyramid network had been employed. The proposed model outperformed the other state-of-art models with F1-score of 95.5%. Wang et al. proposed an MSW management system based on DL and the Internet of Things (IoT) to reduce the cost of waste classification, monitoring, and collection (Wang et al. 2021). Seven different CNN-based models were applied to detect the category of waste. Further, an IoT device was designed to monitor the overall amount of waste produced in the area and the operating state of any waste container via a set of sensors. The MobileNetV3 model surpassed the other six state-of-art employed models by attaining accuracy of 92.24%.

A Multi-Layer Hybrid CNN (MLH-CNN) model which is similar to VGGNet was devised to classify the garbage (Shi et al. 2021). The common optimizers such as Adam, Stochastic Gradient Descent (SGD), and SGDM + Nesterov were compared. Among them, the MLH-CNN classifier along with the SGDM + Nesterov optimizer obtained accuracy of 92.6%. Gyawali et al. employed the ResNet18 model to identify the type of waste, which acquired accuracy of 87.8% (Gyawali et al. 2020). A double fusion method for image-based automatic classification of waste materials was developed (Ahmad, Khan, and Al-Fuqaha 2020). Compared the capabilities of two feature-level fusions and four late fusion methods. The overall performance was improved by 3.5%. Particle Swarm Optimization (PSO) based double fusion performed better than the other employed models with accuracy of 94.58%. A couple of datasets namely VNtrash and TrashNet were used to classify the waste (Vo et al. 2019). The ResNext model was improved by adding two fully connected layers after the global average pooling layer. The proposed model was devised as the DNN-TC model which acquired 98% accuracy for VNTrash and 94% accuracy for TrashNet.

A combination of the pre-trained models such as VGG19, DenseNet169, and NASNetLarge was utilized to predict the type of waste (Huang et al. 2020). The model was demonstrated on a couple of datasets, achieving a classification accuracy of 96.5% and 94%. The types of garbage were classified by using four different DL models (Ruiz et al. 2019). The best classification results were obtained using a combined Inception-ResNet model that achieved 88.6% accuracy. An intelligent waste classification system was constructed with a 50-layer residual net pre-trained model as an extractor and a Support Vector Machine (SVM) as a classifier to categorize the waste (Adedeji and Wang 2019). In addition, the Stochastic Gradient Descent Momentum (SGDM) optimizer was used to accelerate the convergence of the proposed model. Hence, 87% of accuracy was obtained through the proposed model. DenseNet121, DenseNet169, Inception ResNetV2, MobileNet, and Xception models were used to forecast the types of waste classified. Adam and Adadelta optimizers were also utilized. Compared with all the employed models DenseNet121 and InceptionResNetV2 models acquired better results of 95% and 94%, respectively (Aral et al. 2018).

Despite the notable advancements in employing DL models for waste classification in various research studies, there exists a research gap in analyzing larger-sized datasets, specifically focusing on the classification of municipal solid waste. While existing studies have demonstrated promising results in controlled settings and smaller datasets, the scalability and robustness of these models to handle the complexities of larger datasets, particularly in the context of municipal solid waste, remain underexplored. Most of the existing literature tends to concentrate on specific models or limited datasets, often overlooking the challenges associated with the broader scale and diversity of waste materials found in municipal solid waste scenarios. Addressing this gap requires research that delves into the application of DL models on more extensive datasets, providing insights into their effectiveness, generalizability, and potential limitations when dealing with the diverse range of waste categories present in real-world waste management systems. The remaining sections of this paper discuss the Materials and Methods in Section 2, while the Section 3 scrutinizes the results and discussion. Finally, Section 4 concludes the paper.

Materials and methods

The dataset utilized in this study was obtained from Kaggle (Waste Classification Data v2, 2019). It is an image dataset, which consists of 25,474 waste images in total. It was categorized and labeled as organic, recyclable, and non-recyclable as shown in Figure 1. These images were collected from web scraping. Table 1 depicts the distribution of the dataset.

The dataset was pre-processed by fixing the image size to 224, 224 and batch size to 128. The dataset was split into train, valid and test datasets with the ratio of 70, 20, and 10. The data augmentation technique was applied to modify the training dataset by applying various transformations to the existing dataset. This reduces overfitting by exposing the model to more inputs and minimizing its reliance on dataset characteristics and patterns, improving generalization performance. This method creates new variations of the input data, which helps improve the robustness of computer vision models. In this paper, the following transformations were applied such as horizontal and vertical flip, zoom range (0.4), and rotation range (10). This yields the final version of the ORN dataset. The ORN dataset was supplied as an input to the proposed methodology (MSW-Net). The MSW-Net is a hierarchical stacked model with the customized CNN and Bayesian optimized MobileNet as the base model and the Gradient Boosting as the meta-classifier. In all the three classifiers Adam has been used as an optimizer as it delivered the better results. The key performance indicators such as accuracy, precision, recall, F1-Score, and the confusion matrix were utilized to evaluate the performance of the proposed model.

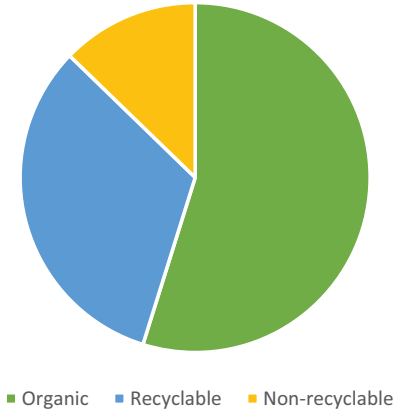


Figure 1. Distribution of types of waste.

Table 1. Summary of the dataset.

S. No.	Category	Data type	Data Count
1	Organic	Image	13,966
2	Recyclable	Image	8,264
3	Non-recyclable	Image	3,244

The proposed MSW-Net model is a hierarchical stacking model. It consists of three layers, including a customized CNN (custom CNN) and Bayesian Optimized MobileNet as the base model and the Gradient Boosting as the meta-classifier as depicted in Figure 2. The decision to utilize MobileNet as the foundational model was motivated by its exceptional performance (Appalanaidu and KumaraVelan 2023; Liu et al. 2023). Similarly, the customized CNN was constructed as a more streamlined iteration of MobileNetV2 (Kaya and Gürsoy 2023; Wang et al. 2023). As the meta-classifier, Gradient Boosting was chosen due to its capability of sequentially combining predictions from numerous weak learners, which are typically decision trees (Emami et al. 2023a, 2023b). The goal of this method is to improve prediction accuracy and overall performance by tweaking model weights iteratively using past mistakes as input. To improve the accuracy of predictions, the hierarchical stacking method is utilized to merge forecasts from multiple models in a stratified structure. Each subsequent-level model uses the predictions of the base model as input features or meta-features. The final prediction is then generated by training a meta-model using the information contained in the meta-feature.

The MSW-Net model features a Customized Convolutional Neural Network (CNN) as its initial layer. It is designed for efficient feature extraction in image classification tasks while balancing computational efficiency and performance. This is inspired by mobile-friendly models, this CNN employs an inverted residual block architecture, combining depthwise and pointwise convolutions. The architecture initiates with a standard 2D convolution, progressing through a series of inverted residual blocks with varying configurations of filters, strides, and expansion factors. Global average pooling condenses spatial dimensions, followed by dense layers with ReLU activation, forming the penultimate feature representation. The final layer is a softmax-activated output layer facilitating multi-class classification.

The custom CNN was implemented on the ORN dataset. It is built with two key building blocks, the convolutional block and inverted residual block functions. The convolutional block allows flexibility between standard and depthwise convolutions, applying batch normalization and ReLU activation. The inverted residual block, inspired by MobileNet-like structures, introduces depthwise separable convolutions for efficient parameter usage and reduced computational cost. The model begins with a standard convolutional layer (16 filters, (3, 3) kernel) with batch normalization and ReLU

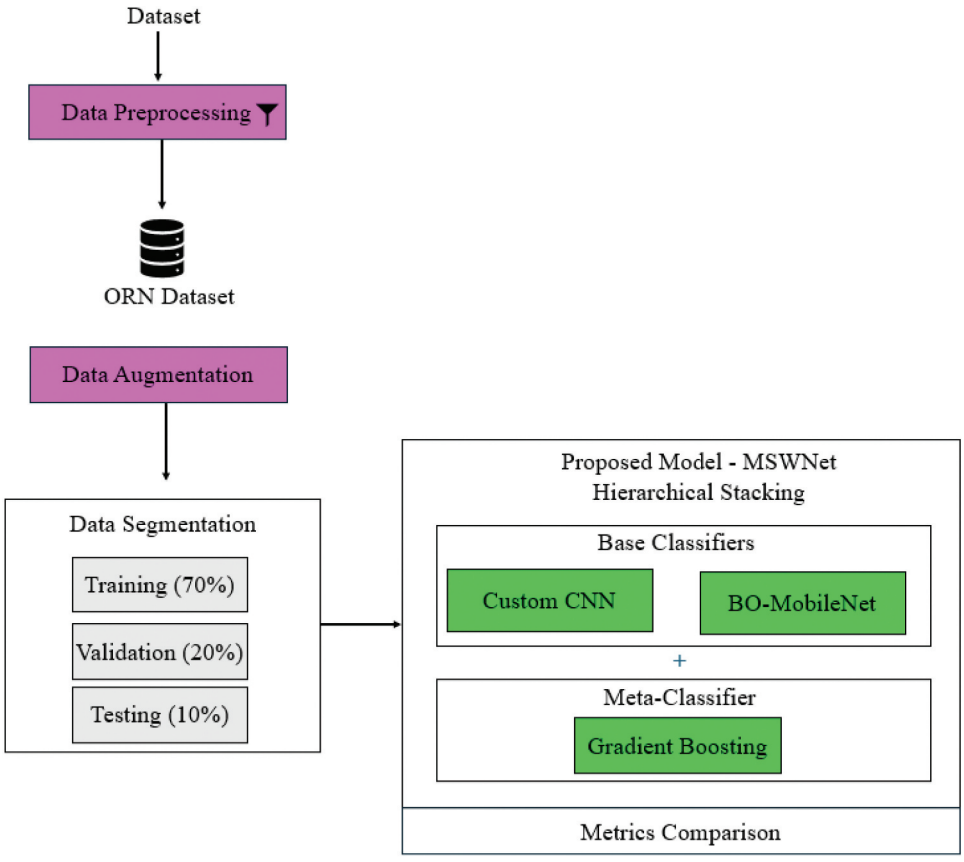


Figure 2. Flow diagram of the proposed model (MSW-Net).

activation. Subsequent inverted residual blocks vary in filter sizes, kernel sizes, and strides to capture hierarchical features. The architecture concludes with global average pooling, a dense layer with ReLU activation, and a final dense layer with softmax activation, producing class probabilities. Refer to Table 2 and Figure 3 for a detailed visualization of the architecture.

The second layer of the MSW-Net incorporates the transfer learning concept using Bayesian Optimized MobileNet (BO-MobileNet). Transfer learning allows the model to leverage pre-trained knowledge from MobileNet, which has been optimized using Bayesian optimization techniques. Leveraging the transfer learning method, the default MobileNet model was implemented on the ORN dataset. The convolutional layers within the model are frozen to prevent further training and retain the pre-trained features. The GlobalAveragePooling2D layer is applied to reduce spatial dimensions, and a Dense layer with softmax activation is added to output probabilities for three classes. To improve the results of

Table 2. Summary of the custom CNN (Base model 1).

Type	Layer	Stride	Filter Shape	Input Size	Output Shape
Input Layer	–	–	(224, 224, 3)	(224, 224, 3)	(224, 224, 3)
Initial Convolution	Conv	(2, 2)	(3, 3, 3, 16)	(224, 224, 3)	(112, 112, 16)
First Block	Depth wise	(1, 1)	(3, 3, 16, 8)	(112, 112, 16)	(112, 112, 8)
Second Block	Depth wise	(1, 1)	(3, 3, 8, 48)	(112, 112, 8)	(112, 112, 48)
	Conv	(2, 2)	(3, 3, 48, 8)	(112, 112, 48)	(56, 56, 8)
Third Block	Depth wise	(1, 1)	(3, 3, 8, 96)	(56, 56, 8)	(56, 56, 96)
	Conv	(2, 2)	(3, 3, 96, 16)	(56, 56, 96)	(28, 28, 16)
Fourth Block	Depth wise	(1, 1)	(3, 3, 16, 96)	(28, 28, 16)	(28, 28, 96)
	Conv	–	(3, 3, 96, 16)	(28, 28, 96)	(28, 28, 16)
	Conv	–	(3, 3, 16, 96)	(28, 28, 16)	(28, 28, 96)
Fifth Block	Depth wise	(1, 1)	(3, 3, 16, 192)	(14, 14, 16)	(14, 14, 192)
	Conv	(2, 2)	(3, 3, 192, 32)	(14, 14, 192)	(7, 7, 32)
Sixth Block	Depth wise	(1, 1)	(3, 3, 32, 192)	(7, 7, 32)	(7, 7, 192)
	Conv	–	(3, 3, 192, 32)	(7, 7, 192)	(7, 7, 32)
	Conv	–	(3, 3, 32, 192)	(7, 7, 32)	(7, 7, 192)
Final Classification	–	–	–	(7, 7, 192)	(3,)

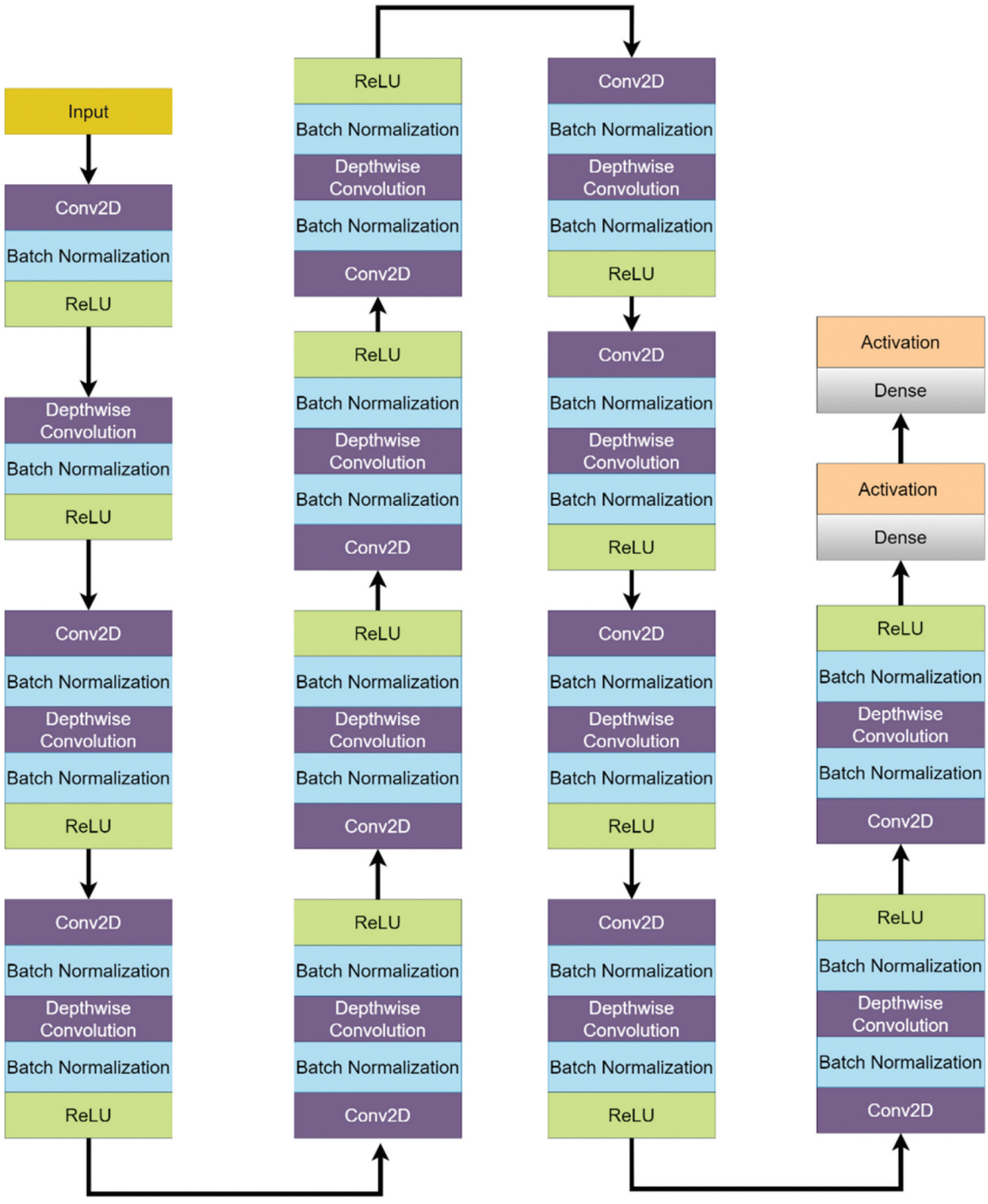


Figure 3. Architecture of custom CNN model.

the pretrained MobileNet model, the hyperparameter tuning was performed using the Bayesian Optimization (BO). The hyperparameters such as number of epochs, size and number of dense layers, dropout rate, batch size, and learning rate were optimized. The result obtained through the Bayesian Optimization is listed in Table 3.

By integrating BO-MobileNet into the architecture, the model benefits from the generalization capabilities of a pre-trained model, enhancing its ability to recognize and classify features in the given dataset. After extracting features through the pre-trained MobileNet layers, a global average pooling layer is implemented to obtain a fixed-size representation of the extracted features. Subsequently, a series of dense layers, determined by the number of dense layers, number of epochs, dropout rate, batch size, and learning rate specified by the optimization techniques, are added. Each dense layer is followed by batch normalization, ReLU activation, and dropout, contributing to the classification process as portrayed in Figure 4 and Table 4. The final dense

Table 3. Results of Bayesian optimized hyperparameters.

Hyperparameters	Range	Outcome
Number of Dense layers	(1, 5)	3
Size of Dense Layer	(32, 1024)	989
Dropout rate	(0.1, 0.3)	0.2
Learning rate	(1e − 5, 1e − 2)	0.0004
Batch Size	(32, 128)	128
Number of epochs	(10, 20)	19

layer with a softmax activation function produces the output probabilities for the different classes, making it the classifier of the model.

The final layer of the MSW-Net is a Gradient Boosting model, which serves as the meta-classifier. Gradient Boosting is a powerful ensemble learning technique that combines the predictions of multiple weak learners to create a robust and accurate final prediction. The Gradient Boosting model is trained on the outputs of the previous two layers, learning to correct any potential errors or misclassifications made by the individual components. This enhances the overall predictive performance of the MSW-Net.

The flow of the proposed model is represented in Figure 5, illustrating the sequential progression of data through the customized CNN, BO-MobileNet, and the Gradient Boosting meta-classifier. This holistic approach ensures the MSW-Net model captures intricate patterns in input data, facilitating accurate multiclass predictions. The overall description demonstrates a smooth transition between layers, minimal grammatical errors, and avoids redundancy.

Results and discussion

Initially, the base classifiers were implemented on the ORN dataset. The custom CNN model, designed by emulating the MobileNetV2 pre-trained model, achieved the following performance metrics. During training, it reached an accuracy of 79%, and precision of 82% while for testing, it obtained an accuracy of 75%, and precision of 78% as mentioned in Table 5. The Custom CNN model, which was built by mimicking the MobileNetV2 pre-trained model, achieved accuracy of 79% in training, 75% in validation and 74% in testing. Later, the MobileNet pre-trained model was applied. It resulted in the accuracy of 92% in training, 90% in validation and 89% in testing.

Further, to improve this model the Bayesian optimization was executed to fine tune the hyperparameters of the MobileNet model (refer to Table 3). The resulting hyperparameters were passed to the base model, which yielded 99% of training accuracy. However, the validation and testing accuracy is only 92%. Gradient boosting is substantially effective due to its superior predicted accuracy,

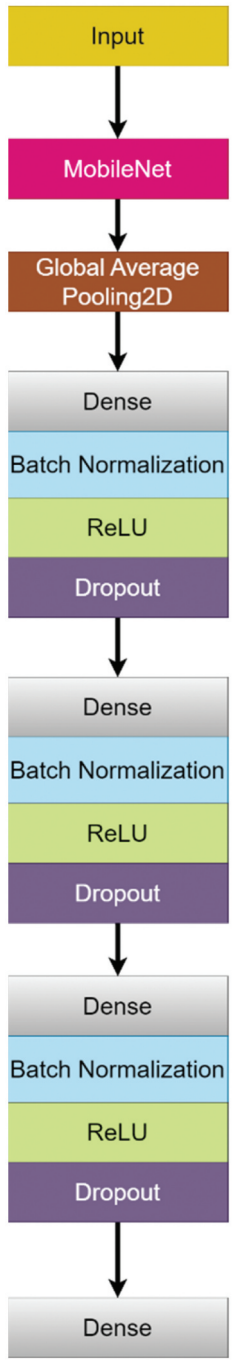


Figure 4. Architecture of Bayesian optimized MobileNet model.

adaptability, resistance to overfitting, and ability to provide valuable insights into feature relevance. This makes it a preferred option for meta-classification tasks in this research. Thus, to further increase the efficiency of the model, a hierarchical stacking model was implemented on the ORN dataset. The BO-MobileNet and Customized CNN act as the base model while the Gradient Boosting Classifier is the meta classifier.

During the training phase, two items of non-recyclables were incorrectly identified as recyclable,

Table 4. Summary of the Bayesian optimized mobileNet model (Base model 2).

Type	Layer	Filter Shape	Input Size	Output Size
Pre-trained	MobileNetV2	–	(img_size[0], img_size[1], 3)	(variable, variable, 1280)
Trainable	GlobalAveragePooling2D	–	(variable, variable, 1280)	(1, 1, 1280)
Trainable	Dense	(988.91,)	(1, 1, 1280)	(1, 1, 988.91)
Trainable	BatchNormalization	–	(1, 1, 988.91)	(1, 1, 988.91)
Trainable	Activation (ReLU)	–	(1, 1, 988.91)	(1, 1, 988.91)
Trainable	Dropout	–	(1, 1, 988.91)	(1, 1, 988.91)
Trainable	Dense	(988.91,)	(1, 1, 988.91)	(1, 1, 988.91)
Trainable	BatchNormalization	–	(1, 1, 988.91)	(1, 1, 988.91)
Trainable	Activation (ReLU)	–	(1, 1, 988.91)	(1, 1, 988.91)
Trainable	Dropout	–	(1, 1, 988.91)	(1, 1, 988.91)
Trainable	Dense	(988.91,)	(1, 1, 988.91)	(1, 1, 988.91)
Trainable	BatchNormalization	–	(1, 1, 988.91)	(1, 1, 988.91)
Trainable	Activation (ReLU)	–	(1, 1, 988.91)	(1, 1, 988.91)
Trainable	Dropout	–	(1, 1, 988.91)	(1, 1, 988.91)
Trainable	Dense	(3,)	(1, 1, 988.91)	(1, 1, 3)

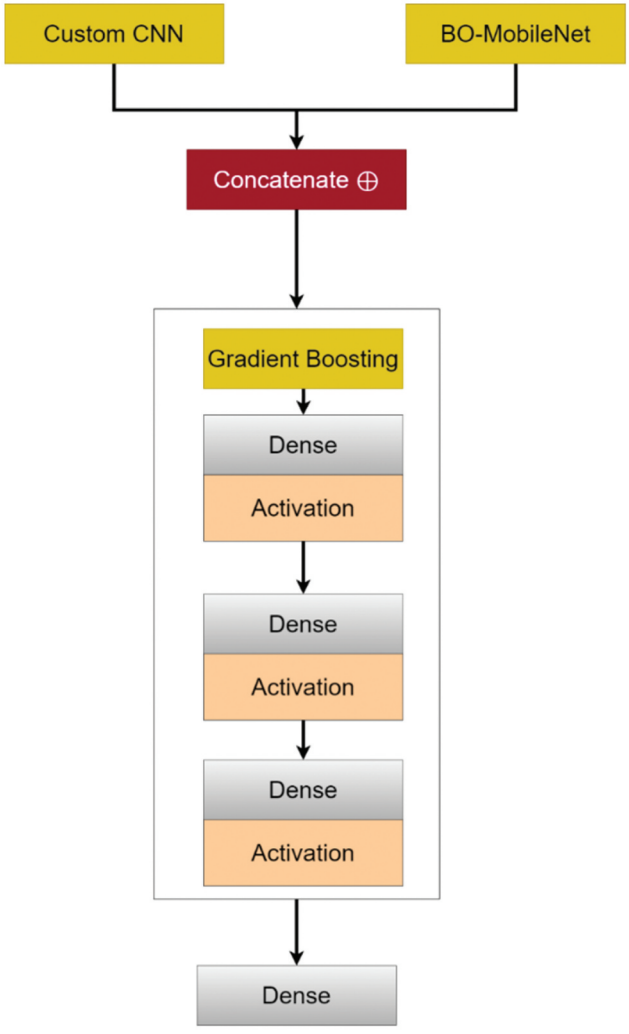


Figure 5. Architecture of MSW-Net model.

while one was erroneously classified as organic as shown in Figure 6. This demonstrates that class 0 has effectively performed by accurately identifying the objects. Nevertheless, the incorrect categorization of non-recyclable items as recyclable and organic materials increased to 35 and 65, respectively, in the validation dataset (refer to Figure 7). Despite a decline in validation accuracy during the testing phase, it ultimately rebounded to achieve the highest level of accuracy, with only 26 misclassified items. Considering organic waste, the training of data was executed proficiently without any inaccurate categorizations. However, there were 98 misclassified objects in the validation dataset. During the testing phase, it demonstrated a 99% accuracy rate, with only 18 out of 1384 items being misclassified. Despite a single mistake, the final class recyclable was also classified with the highest outcome. Similarly, it achieved peak accuracy levels during the testing phases as depicted in Figure 8. The MSW-Net model obtained accuracy score of 96.43%, precision score of 96.42%, recall score of 96.43%, and F1 score of 96.42% in the testing datasets, respectively. These scores collectively demonstrate the model's high effectiveness in accurately classifying various types of waste, with minimal false positives and false negatives. Hence, the proposed methodology outperforms the other basic models.

The Table 6 provides an evaluation of MSW-Net's efficacy in comparison to other contemporary models. In particular, MSW-Net achieved 96.43% of accuracy, 96.33% of precision, 96.43% of recall and 96.32% of F1-score, which is higher than all of the other models mentioned. These metrics surpass those of all other models mentioned, indicating its superior capability in accurately distinguishing various waste categories. The high precision and recall values suggest that MSW-Net is reliable in both recognizing relevant instances and minimizing false positives and negatives, making it an efficient model for accurate waste classification. The runtime efficiency of its training, validation, and testing phases is particularly significant, consuming a total of 21, 7, and 4 seconds, respectively. This extensive

Table 5. Results of the employed models.

		Custom CNN	MobileNet	BO-MobileNet	MSW-Net
Training	Accuracy (%)	79	92	99	99
	Precision (%)	82	92	99	99
	Recall (%)	79	92	99	99
	F1-Score (%)	79	92	99	99
Validation	Accuracy (%)	75	90	92	95
	Precision (%)	78	90	92	95
	Recall (%)	76	90	92	95
	F1-Score (%)	75	90	92	95
Testing	Accuracy (%)	75	78	91	96.43
	Precision (%)	78	79	91	96.42
	Recall (%)	75	78	91	96.43
	F1-Score (%)	74	77	91	96.42

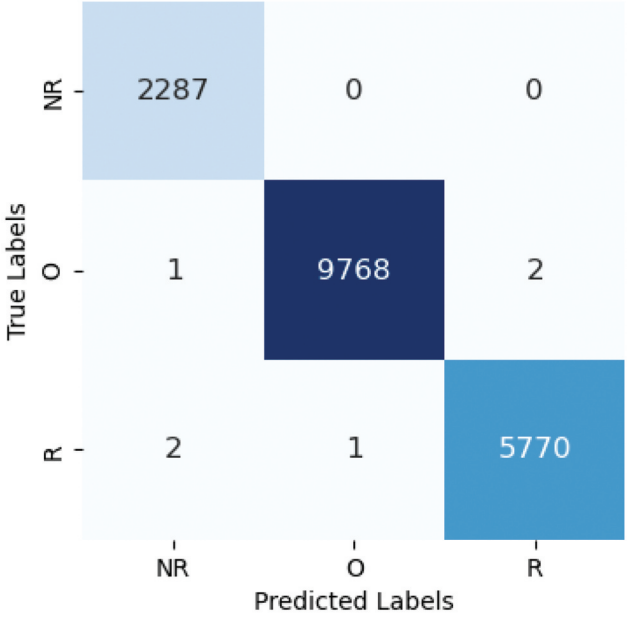


Figure 6. Confusion matrix (Training).

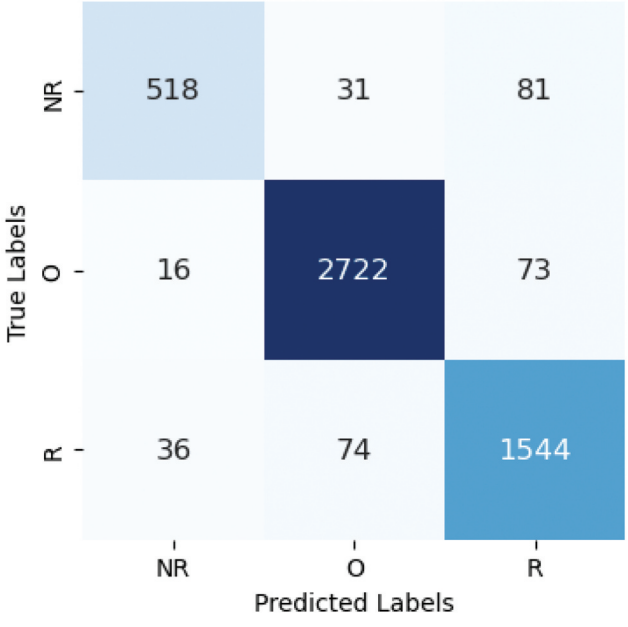


Figure 7. Confusion matrix (Validation).

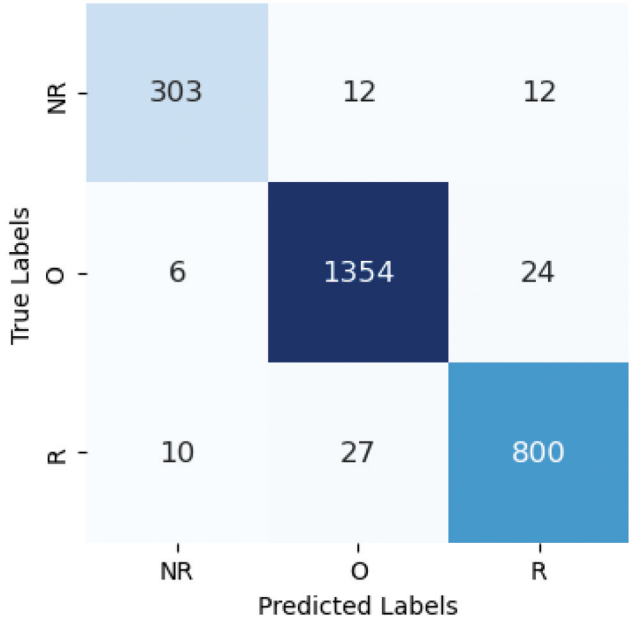


Figure 8. Confusion matrix (Testing).

comparison emphasizes the higher performance of the MSW-Net in waste categorization tasks, suggesting it could be a good fit for waste management procedures.

The MSW-Net performs well as the hierarchical stacking approach combines the strengths of base models (Custom CNN and BO-MobileNet) with a robust meta-classifier (Gradient Boosting), which leads to accurate classification. Additionally, the use of Bayesian optimization for hyperparameter tuning enhances model performance by fine-tuning parameters for optimal results. In general, these factors contribute to the exceptional performance of MSW-Net in the precise classification of MSW. The utilization of MSW-Net by local authorities for waste classification represents a substantial progression in the preliminary phases of waste management methods. To improve the efficiency and efficacy of trash sorting, MSW-Net stands out as a possible option due to its remarkable accuracy and efficient runtime.

Table 6. Comparison of the proposed methodology with the existing study.

S. No.	Models	Testing dataset Accuracy
1	MobileNetV1 (Nurahmadan et al. 2021)	88.26%
2	MobileNet (Thanawala, Sarin, and Verma 2020)	92.65%
3	Weighted average ensembled of InceptionV3, VGG16, ResNet152V2, MobileNetV2 and DenseNet201 (Gupta and Mukherjee 2022)	93.62%
4	Custom InceptionNetV3 (Loganayagi and Usha 2023)	94%
5	Proposed model – MSW-Net	96.43%

Conclusion

Every conceivable human action produces waste, commencing with the extraction of raw materials and culminating in the manufacturing of the final product. Even though the MSW only accounts for around 10% of the overall waste generated, treatment and management of the same frequently account for over one-third of public sector funding for pollution abatement and control. Soil and water contamination, air pollution, climate change, improper land use, and unsightly landscapes are among ways in which improper waste management endangers human and environmental health. To address the challenges of managing municipal solid waste, this study introduces the MSW-Net model, a computer vision-based DL approach for waste classification using an image dataset. The MSW-Net is a stacking model with custom CNN and BO-MobileNet as a base classifier and gradient boosting as the meta-classifier. The MSW-Net model outstripped the other existing models with the testing accuracy, precision, recall and F1-score of 96.43%, 96.42%, 96.43% and 96.42% respectively. Thus, as a future work, this model would be implemented with a dataset with more municipal solid classes. Further, it could be integrated into an IoT-based device and a miniature robot would be designed to automatically classify the waste.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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Data availability statement

The dataset generated and/or analyzed during the current study is available from the corresponding author on reasonable request.

Author contributions

Vaishnavi Jayaraman: Conceptualization, Methodology, Data Curation, Visualization, Investigation, Software and Writing – Original draft preparation.
Arun Raj Lakshminarayanan: Supervision, Validation, Reviewing and Editing.

Code availability

The code generated during the current study is available from the corresponding author on reasonable request.

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